

Diphenylguanidine



What is it?

Diphenylguanidine (DPG) is a rubber accelerator. A rubber accelerator is a chemical used when making rubber. It speeds up the liquid to solid rubber transformation. It can become a part of the final product in this process. This may cause allergic contact dermatitis. Note that this is different from a latex allergy.

DPG may be in any type of rubber product. It may also be in other chemicals such as germicides and pesticides. DPG is also a complexing agent used in the detection of metals and organic bases.

DIPHENYLGUANIDINE can also be known as:

- 1,3-Diphenylguanidine
- N,N-Diphenylguanidine
- DPG, DPG Accelerator
- Nocceler D
- Vulkazit, Vulkacit D/C
- Sanceler D
- Vulcacid D
- Melaniline
- Soxinol D
- Sym-diphenylguanidine

Some Possible Uses of DIPHENYLGUANIDINE

/ Where it might be found:

- Rubber and nitrile gloves
- Rubber shoes, insoles, soles
- Rubber hip waders, gators
- Elastic in clothing, socks, and girdles
- Support stockings
- Non-slip waist bands, elastics
- Brassiere cups, underwear, or panties
- Feminine hygiene products
- Swimwear, swim cap, goggles, sauna suits

- Wet suit, diving gear
- Rubber pillow, mattress, or sheet
- Makeup applicator, lash curler
- Ear plugs, headphones
- Hairbrush handle, hairnet
- Toothbrush handle, dentures, dental dams
- Baby bottle nipples, pacifiers
- Balloons, balls, toys
- Medical devices (stethoscope, IV port, tourniquet, BP cuff, CPAP strap, tubing)
- Dialysis equipment
- Braces and supports
- Safety aprons, goggles
- Respirators, masks
- Microscope eyepiece
- Rubber prosthesis and splint
- Rubber dropper
- Bunion pad
- Condom or diaphragm
- Rubber vibrator
- Phone armband
- Pen handle, pencil grip, eraser
- Mouse pad, wrist support
- Glue, envelopes, stamps
- Rubber cement, Rubber bands
- Kitchen utensil rubber handle
- Rubber mats, carpet backing, underlay pad
- Racquet and club handles
- Hockey pads, soccer shin pads
- Foam shoulder pad
- Rubber hose, tubing, seals, cables
- Steering wheel, bike handle
- Auto hoses, gaskets, fan belts
- Shock absorbers, springs, tires, inner tubes, Fuel tank lining
- Duct tape, Electrical cords, tool handles
- Weather stripping
- Conveyor belt
- Synthetic resins in textile/dye industry

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Diphenylguanidine



- Some, but not all, Neoprene (polychloroprene)

Other products that might react like

DIPHENYLGUANIDINE:

Carba Mix

What should I do to avoid

DIPHENYLGUANIDINE (DPG)?

Avoid ALL rubber, which may contain DPG.

Replace Rubber Items with these Alternatives:

- Acrylic
- Buttons (instead of elastic)
- Fabric
- Mylar
- Plastic
- Polyethylene foam
- Silicone
- Vinyl (AKA polyvinyl chloride, PVC)
- Wood

Use VINYL, non-latex gloves when you clean the house or wash dishes. Wear heavy duty vinyl gloves to protect against workplace contact.

Elastic waistbands and underwear with elastic may contain rubber. Choose underwear that does not have an elastic waistband, or buy all-cotton clothing (cottonique.com, rawganique.com, decentexposures.com; make sure rubber-free).

Ask your provider for other options if your products or items may contain diphenylguanidine.

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Which gloves may NOT have diphenylguanidine?

- Household use:
 - Vinyl, non-latex gloves
- Medical Examination (non-sterile) gloves:
 - Ansell Micro-Touch NitraFree
 - Cardinal Health Low Dermatitis Potential Nitrile
 - Medline SensiCare Powder-Free
 - Sempermed SemperSure Nitrile
 - Showa N-Dex Accelerator-free Nitrile
 - SmartPractice Reflection Sky Sensitive Nitrile
- Surgical (sterile) gloves:
 - Ansell GAMMEX Non-Latex Neoprene Surgical
 - Cardinal Health Protexis Neoprene Essential Surgical
 - Medline DermAssure Green Powder-Free Neoprene Surgical
 - Mölnlycke Health Biogel NeoDerm Surgical
- Chemical resistant gloves:
 - Honeywell Safety Silvershield/4H chemically resistant

For more rubber-accelerator free glove options: *Dermatitis* 2018;29:66; *Dermatitis* 2018;29:13.

For more rubber-free options: *Dermatitis* 2019;30:87.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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